MSD MANUAL
Professional Version

Overview of Cutaneous Porphyrias

Full Review: Jan 2025 By ***Herbert L. Bonkovsky****, MD, Wake Forest University School of Medicine | **Sean R. Rudnick**, MD, Wake Forest University School of Medicine | Peer reviewed by **Glenn D. Braunstein**, MD, Cedars-Sinai Medical Center*
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Cutaneous porphyrias result from deficiency (and in one case, excess) of certain enzymes in the heme biosynthetic pathway (see table Substrates and Enzymes of the Heme Biosynthetic Pathway), resulting in a relatively steady production of phototoxic porphyrins in the liver or bone marrow. These porphyrins accumulate in the skin and, on sunlight exposure (visible light, including near-ultraviolet [UV]), generate cytotoxic radicals that cause recurrent or unremitting cutaneous manifestations.

(See also Overview of Porphyrias.)

Cutaneous porphyrias include

- Porphyria cutanea tarda (PCT)
- Erythropoietic protoporphyria (EPP)
- Congenital erythropoietic porphyria (CEP—see table Some Less Common Porphyrias)
- X-linked protoporphyria (XLPP), sometimes regarded as a clinical variant of erythropoietic protoporphyria
- Hepatoerythropoietic porphyria (HEP—see table Some Less Common Porphyrias), sometimes regarded as a type of PCT (extremely rare)

The acute porphyrias variegate porphyria (VP) and hereditary coproporphyria (HCP) may also have cutaneous manifestations.

In all cutaneous porphyrias except erythropoietic protoporphyria and X-linked protoporphyria, cutaneous photosensitivity manifests as fragile skin and bullous eruptions. Skin changes generally occur on sun-exposed areas (eg, face, neck, dorsal aspects of hands and forearms) or traumatized skin. The cutaneous reaction is insidious, and often patients are unaware of the connection to sun exposure. In contrast, the photosensitivity in erythropoietic protoporphyria and X-linked protoporphyria occurs within minutes or hours after sun exposure, manifesting as a burning pain that persists for hours, without any blistering and often without any objective signs on the skin.

However, swelling and erythema may occur. With chronic exposure, lichenification of the skin may develop, usually most noticeable over the knuckles of the hands and on the face.

Chronic liver disorders are common in cutaneous porphyrias.

The cutaneous porphyrias are all accompanied by elevated total plasma porphyrins, and are specifically diagnosed by measurements of porphyrins in red blood cells, plasma, urine, and stool, as well as by genetic or enzyme analysis. Treatment involves avoidance of sunlight, measures to protect the skin, and sometimes other treatments directed to the specific diagnosis.

TABLE

Some Less Common Porphyrias

Description	Symptoms and Signs	Diagnostic Approach	Treatment
Congenital erythropoietic porphyria (Günther disease)			
Severe deficiency of uroporphyrinogen III cosynthase (UROS)	In utero or shortly after birth: Severe cases manifesting as nonimmune hydrops Soon after birth: Skin blistering, hemolytic anemia, hyperbilirubinemia, red urine, dark diapers that show a red fluorescence under UV light Phototherapy for hyperbilirubinemia leads to severe skin blistering	Porphyrins in plasma, urine, and stool elevated to levels higher than those in other porphyrias, with uroporphyrin I and coproporphyrin I the predominant porphyrins in urine and stool Urinary ALA and PBG virtually normal Can be confirmed by low RBC UROS activity (< 10%), but test	Avoidance of sunlight (including lights for treating neonatal hyperbilirubinemia) Use of sun-protective clothing Iron reduction therapy Avoidance of skin trauma Prompt treatment of secondary bacterial infections to help prevent scarring Splenectomy possibly beneficial for

Description	Symptoms and Signs	Diagnostic Approach	Treatment
	In adulthood: Facial disfiguration, increased hair growth, corneal scarring (possibly severe), hemolytic anemia, splenomegaly, erythrodontia, deposition of porphyrins in bone, bone demineralization (possibly substantial)	not readily available Genetic analysis of UROS gene, which reveals homozygous or compound heterozygous mutations on chromosome 10 (most common mutation is pC73R) For *in utero* diagnosis: Measurement of amniotic porphyrins or genetic analysis	patients with hemolytic anemia Repeated RBC transfusions and hydroxyurea to keep bone marrow porphyrin production low; deferoxamine for transfusion-related iron overload Hematopoietic stem cell transplantation potentially curative
Hepatoerythropoietic porphyria			
Severe deficiency of uroporphyrinogen decarboxylase (UROD)	Early childhood Skin blistering Red urine Anemia	Elevated isocoproporphyrin in stool and urine Elevated zinc protoporphyrin in RBCs (to differentiate from PCT) Confirmed by very low RBC UROD activity Genetic analysis of UROD gene,	Avoidance of sunlight Phlebotomy possibly beneficial to patients with milder cases Treatment of severe disease similar to that of congenital erythropoietic porphyria

Description	Symptoms and Signs	Diagnostic Approach	Treatment
		which reveals homozygous or compound heterozygous mutations	
Dual porphyria			
Disorders resulting from deficiencies of 2 enzymes of the heme biosynthetic pathway	Clinical and biochemical manifestations of both disorders In acute porphyrias: Neurovisceral symptoms triggered by porphyrogenic agents In cutaneous porphyrias: Hypersensitivity to sunlight with blistering and fragile skin	Porphyrin and porphyrin precursor excretion patterns Confirmed by family history and enzyme analyses	In acute porphyrias: Avoidance of triggering agents In cutaneous porphyrias: Skin protection and avoidance of sunlight

ALA = delta-aminolevulinic acid; PBG = porphobilinogen; PCT = porphyria cutanea tarda; RBC = red blood cell; UV = ultraviolet.

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